

can purchase online and make online bill and taxi fare payments from their mobile phone.

Mobipay is established on a single technology standard supplied by Atos Origin, which is a renowned international information technology services company that deals with consulting, system integration and managed operations. Atos Origin performs telcofinance activities that involve banks, payment processors, merchants and mobile network operators. The payment system first validates the payment request made from the user's mobile handset against credit/debit cards, virtual wallets, billing systems, etc and then routes authorisation requests for transactions to respective parties. Apart from providing payment services, Atos Origin also provides managed services in business process outsourcing and infrastructure solutions.

Mobipay systems are comparatively low-cost and low-risk. The systems are designed for end-to-end control by trusted partners and USSD messaging supports secure delivery of sensitive information to authorised users.

Reliance mPay. Reliance mPay is a virtual credit card, jointly launched by HDFC bank and Reliance Communications in India, that allows users to pay from their Reliance mobile phone. In order to avail this service, the users must have an HDFC bank credit card and a Reliance mobile phone. They don't have to carry the physical credit card, as the mobile phone acts as the credit card to make postpaid bill payment, book movie or travel tickets, pay electricity bills, make payments in shops and restaurants and shop on the Internet.

The customers are issued a PIN for using the virtual credit card. To make payments, they need to enter the PIN and all the transactions must be initiated from the registered mobile number to which the mPay service PIN has been issued. The mPay credit card can be used with HDFC Bank and any merchant establishment that accepts

payment via Reliance mPay. The payment is automatically debited from the user's credit card account. Users don't need to enter a credit card number, card verification information or expiry date to use the mPay service. HDFC bank would authenticate all the information and then initiate and authorise all payments using the customer's mobile handset.

With a virtual credit card, there is less chance of misuse, as at the time of payment, only the PIN is to be entered and not the credit card number. So in case your mobile phone is stolen or lost, the credit card number cannot be misused as the mobile phone does not contain the credit card number. Owing to their convenience and safety, virtual credit cards have become a preferred medium of transaction.

SEMOPS. Secure mobile payment service (SEMOPS) is a project to develop a secure and universal mobile payment service that can be used uniformly in all kinds of mobile commerce/electronic commerce transactions. SEMOPS technology allows multichannel access (including Web, GPRS, GSM and SMS) to bank accounts and also electronic/mobile payment transactions between customers and merchants irrespective of whether they are banking with the same organisation or not. Based on this concept, a truly global payment system can be developed with addition of more and more banks.

With SEMOPS, consumers will be able to perform different types of transactions including point-of-sale purchases, person-to-person money transfer, bill payment, payment at vending machines, mobile phone bill payment, parking fee payment and online purchase through the Internet. It provides a comprehensive mobile payment solution that can be easily adopted by different banks, merchants, mobile vendors and consumers. Users will be able to pay for anything, anywhere and at any time.

SEMOPS is designed to follow standard banking processes so that

financial institutions can easily integrate it into their existing infrastructure. It follows a multi-layer security architecture where the consumer has full control of the payment process and approves each and every stage of the transaction with the help of the accompanying PIN number. If the mobile phone is lost, it cannot be used for making purchase without the appropriate PIN number.

SEMOPS was first introduced in Hungary and then spread over other European countries. It is open to banks or mobile operators who wish to join and offer services to their existing client base in a universal manner.

Future outlook

With the explosive growth in mobile technology and rapid penetration of mobile phones in developing countries, the future holds immense possibilities for mobile payments. NFC-based contactless smart cards will enable users to combine the functionalities of credit card, debit card, bank account card as well as personal identification card in a single device in the form of a mobile phone. The mobile phone will replace the pocket money and provide a low-cost, low-risk alternative to credit/debit cards for making payments anytime, anywhere and for anything.

However, in order to make mobile payment universal and acceptable to all merchants, banks and mobile operators, proper standardisation is needed. Also, the success of any mobile payment system largely depends on consumer acceptance. The consumers will adopt any new mobile payment system if they find it convenient, easy to use and secure. So future developments in mobile payment system must give special emphasis on security and cost factor. ●

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